

12(a) Find the minimum cost spanning tree of a given undirected graph using prim's algorithm.

```
#include <stdio.h>
#include <limits.h>
```

```
#define MAX 100
```

```
int minKey (int key[], int mstSet[], int v) {
```

```
    int min = INT_MAX, min_index;
```

```
    for (int v = 0; v < V; v++) {
```

```
        if (mstSet[v] == 0 && key[v] < min) {
```

```
            min = key[v];
```

```
            min_index = v;
```

```
        }
```

```
    }
```

```
    return min_index;
```

```
}
```

```
void printMST (int parent[], int graph[MAX][MAX], int V,
```

```
int totalCost = 0;
```

```
printf ("\n Edges in the Minimum Spanning Tree: \n");
```

```
for (int i = 1; i < V; i++) {
```

```
    printf ("%d -- %d == %d \n", parent[i],
```

```
    i, graph[i][parent[i]]);
```

```
    totalCost += graph[i][parent[i]];
}
```

```
printf ("Total cost of MST = %d \n", totalCost);
```

```
}
```

```

void primMST(int graph[MAX][MAX], int V) {
    int parent[V];
    int key[V];
    int mstSet[V];

    for (int i = 0; i < V; i++) {
        key[i] = INT_MAX;
        mstSet[i] = 0;
    }

    key[0] = 0;
    parent[0] = -1;

    for (int count = 0; count < V - 1; count++) {
        int u = minKey(key, mstSet, V);
        mstSet[u] = 1;

        for (int v = 0; v < V; v++) {
            if (graph[u][v] && mstSet[v] == 0 &&
                graph[u][v] < key[v]) {
                parent[v] = u;
                key[v] = graph[u][v];
            }
        }
    }

    printMST(parent, graph, V);
}

int main() {
    int V;
    int graph[MAX][MAX];
    printf("Enter number of vertices:");
    scanf("%d", &V);

```



```

printf("Enter adjacency matrix :\n");
for (int i=0; i<V; i++) {
    for (int j=0; j<V; j++) {
        scanf("%d", &graph[i][j]);
    }
}

primeMST (graph, V);

return 0;
}

```

output;

Enter number of vertices: 5

Enter adjacency matrix:

0 2 0 6 0

2 0 3 8 5

0 3 0 0 7

6 8 0 0 9

0 5 7 9 0

Edges in minimum Spanning tree:

0 -- 1 == 2

1 -- 2 == 3

0 -- 3 == 6

1 -- 4 == 5

Total cost of MST = 16.